Review of Signals and Systems

AI 321

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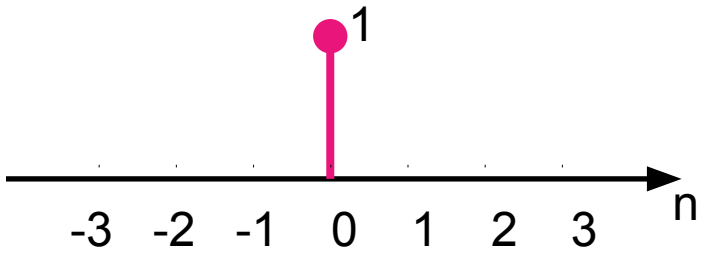
Discrete Time Signal Representation

- Most of the discrete time signals we encounter can be described by their wave characteristics (e.g. amplitude, frequency, phase, decay time)

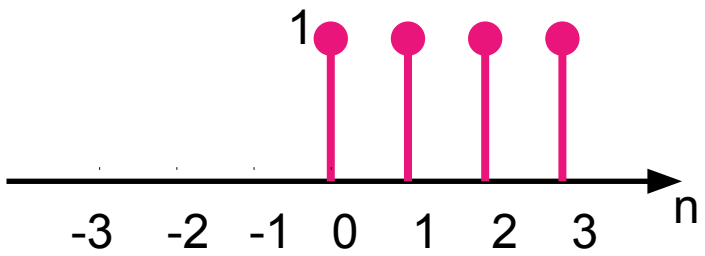
- In general, they can be mathematically represented using **elementary functions, exponential functions,** and **sinusoidal functions**

Elementary Functions

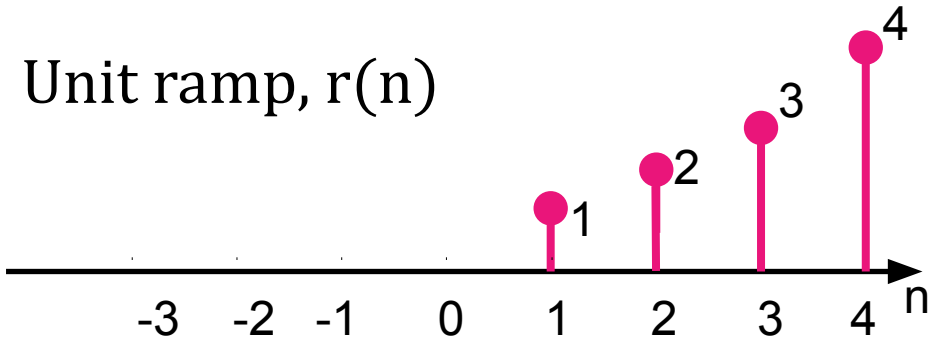
Unit impulse, δ(n)



Unit step, u(n)



Unit ramp, r(n)



Exponential Functions

Exponential, $a^n u(n)$

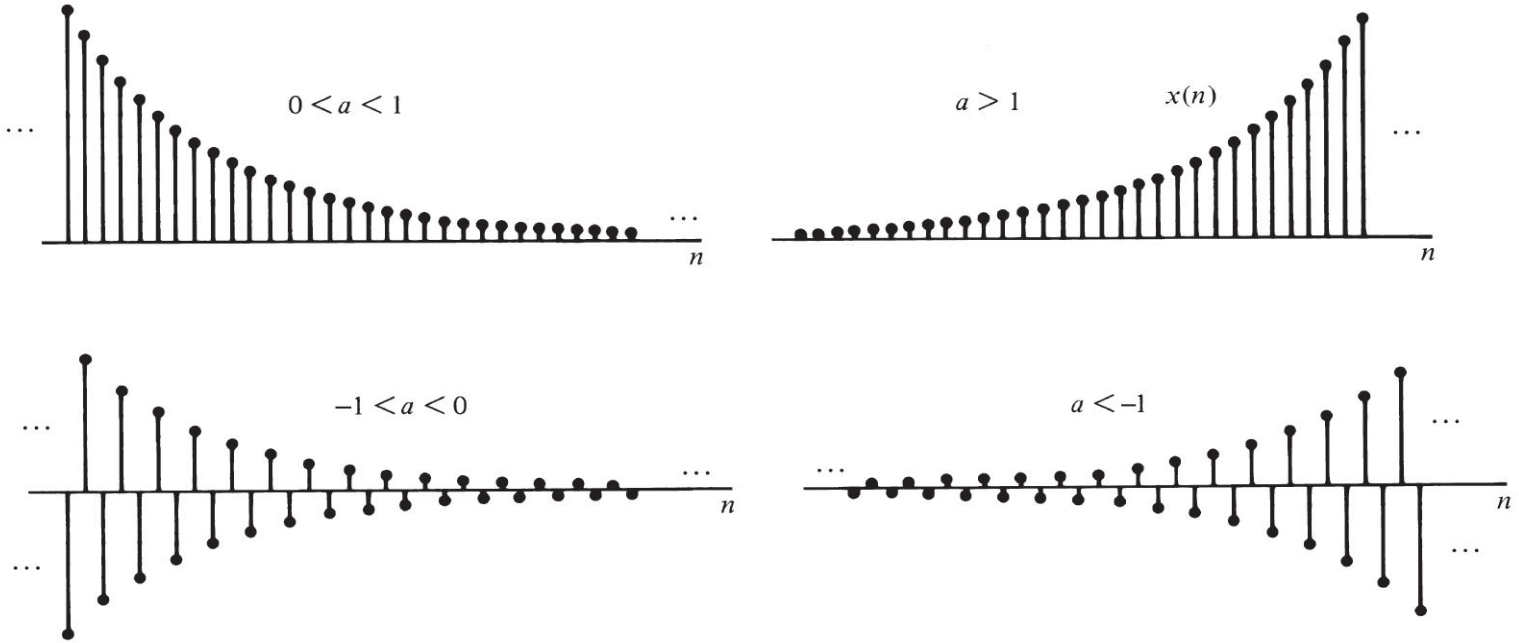


Figure 2.1.5 Graphical representation of exponential signals.

Sinusoidal Functions

$$x[n] = \text{Asin}(\omega n + \emptyset)$$

$$x[n] = \text{Asin}\left(\frac{2\pi kn}{N} + \emptyset\right)$$

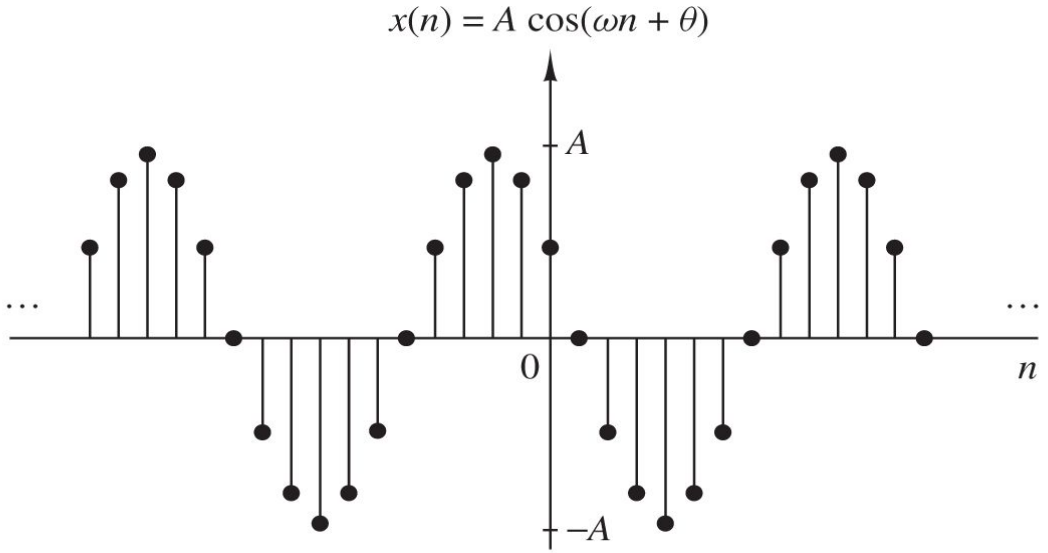


Figure 1.3.3 Example of a discrete-time sinusoidal signal ($\omega = \pi/6$ and $\theta = \pi/3$).

Complex Exponential signals

- In $x(n) = a^n$, if a is complex, that is $a = re^{j\omega_0}, r > 0, \omega \neq 0, \pi$, then $x(n)$ is complex-valued,

$$x(n) = r^n e^{j\omega_0 n} = x_R(n) + jx_I(n)$$

where $x_R(n) = r^n \cos(\omega_0 n)$ and $x_I(n) = r^n \sin(\omega_0 n)$ are the real and imaginary components

What happens if $r < 1$? if $r > 1$?

Combining basic signals with addition, multiplication and time-shift

Try to visualize (plot) the following: signals

- $2u(n) - 2u(n-5)$
- $\delta(n+1) + 2\delta(n-1) - 0.5\delta(n-2)$
- $5(\frac{1}{2})^n u(n)$
- $-3u(-n-1)$
- $(\frac{1}{2})^n u(-n-1)$

Signal Frequency vs. Sampling Frequency

Nyquist's Sampling Theorem:

" If the highest frequency contained in an analog signal, $x_a(t)$, is F_{max} and the signal is sampled at a rate $F_s > 2F_{max}$, then $x_a(t)$ can be exactly recovered from its sample values using the interpolation function"

- Not following Nyquist criterion may lead to **aliasing**

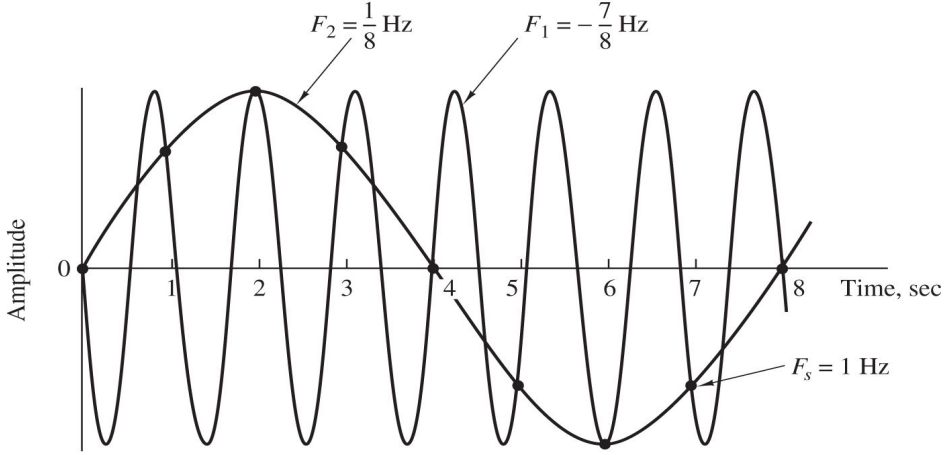
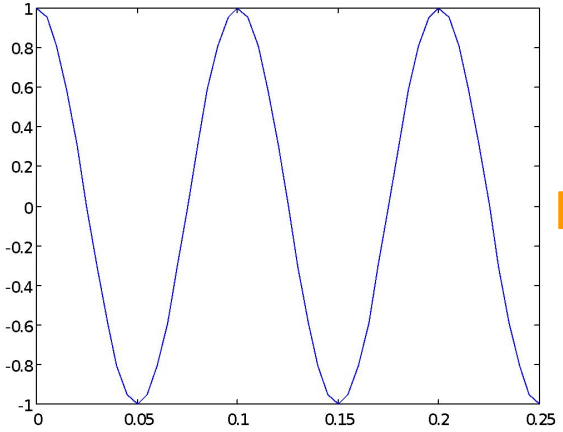
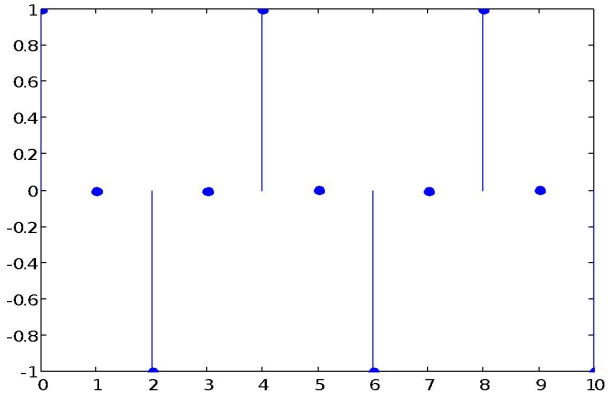


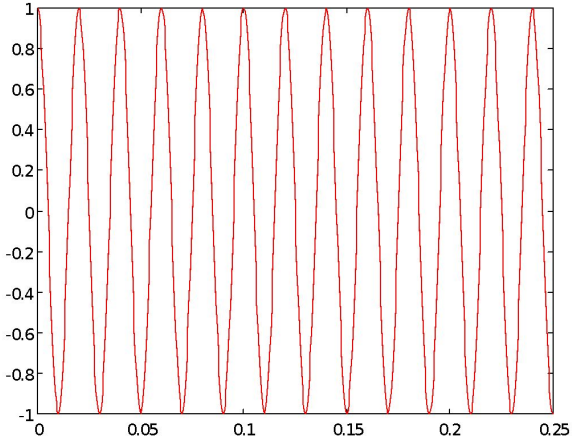
Figure 1.4.5 Illustration of aliasing.

Signal Frequency vs. Sampling Frequency

$F_s = 40$ Hz







$x_1(t) = cos(2\pi\ 10t)$

$x(n) = cos(0.5\pi n)$
$= cos(2.5\pi n)$

$x_2(t) = cos(2\pi\ 50t)$

Signal Classification

- The *energy* of a signal $x(n)$ is given as:

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 \geq 0$$

- Average power or simply the *power* of the signal defined as:

$$P_x = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n|^2 \geq 0$$

signals with finite energy are *energy* signals, signals with finite power are *power* signals. Energy signals have zero power, power signals have infinite energy.

Signal Classification

- DT-signal is *periodic* if $x(n+N) = x(n)$, where N is the *fundamental period*. Otherwise the signals is *aperiodic*.
  - Complex exponentials are periodic if and only if $\omega_0/(2\pi) = k/N$, rational.
  - Periodic signals are power signals with P:

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$

- Finite duration: $x(n) = 0$, for $n < N_1$ and $n > N_2$, where N_1 and N_2 are finite integers and $N_1 \leq N_2$. If $N_1 = -\infty$ and $N_2 = \infty$ the signal is *infinite* duration
- Causal: if $X(n) = 0$ for $N < 0$, otherwise *noncausal*.
- Real-valued signals are *even* if $x(-n) = x(n)$ and *odd* if $x(-n) = -x(n)$.

Discrete Time Signal Classification

- The unit step has infinite energy, it is a power signal with $P=0.5$
- Complex exponential $Ae^{j\omega n}$ is a power signal with $P=A^2$
- The unit ramp sequence is neither a power signal nor an energy signal

Discrete Time Signal Classification

A signal *x(n)* is periodic with period *N* if and only if

$$x(n+N)=x(n) \quad \textit{for all } n$$

Ex. $x(n)=A\sin(2\pi f_0 n)$ periodic when f_o is a rational number

➢ **Periodic** signals are **power** signals.

Discrete Time Signal Classification

A DT signal *x(n)* is **even** if:

$$x[n] = x[-n]$$

A DT signal *x(n)* is **odd** if:

$$x[n] = -x[-n]$$

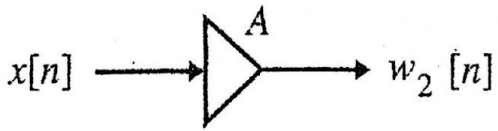
Summary of signal representation

- Discrete time signals can be obtained from sampling and quantizing continuous time signals.
- To ensure proper signal representation, Nyquist criteria and proper quantization must be followed
- Elementary functions, exponential functions, and sinusoidal functions can be used to mathematically represent any discrete time signal
- Discrete time signals can be classified as power or energy, periodic or aperiodic, and odd or even

Discrete Time Signal Manipulation

Amplitude scaling & inversion

$$x[n] \rightarrow Ax[n]$$



$$w_2[n] = Ax[n]$$

$|A| > 0 \rightarrow$ amplify/boost

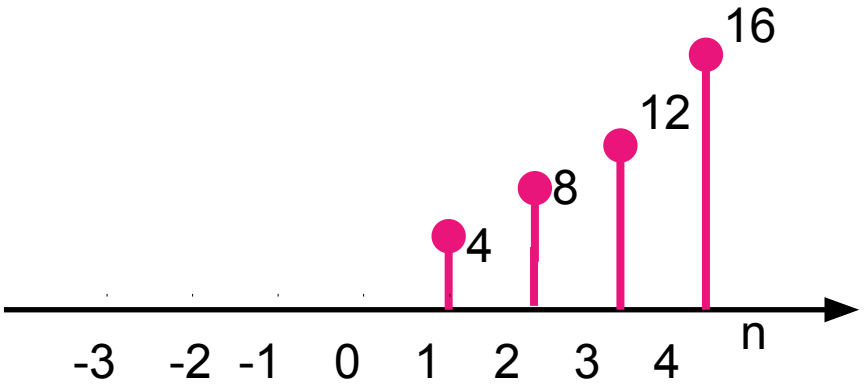
$0 < |A| < 1 \rightarrow$ gain reduction/attenuate

$A < 0 \rightarrow$ invert

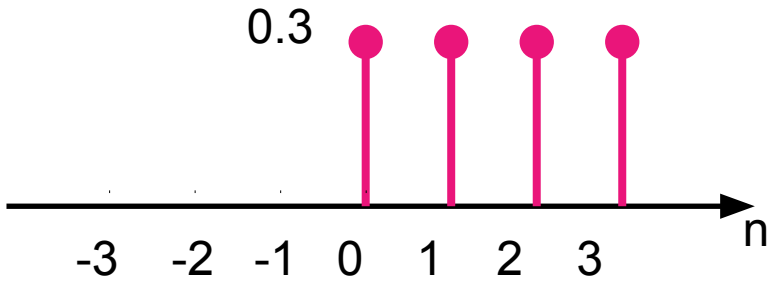
Discrete Time Signal Manipulation

Amplitude scaling

Amplified ramp, *4r(n)*



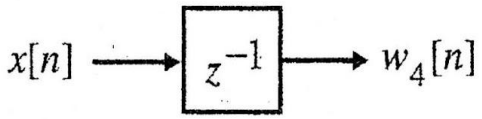
Attenuated step, *0.3u(n)*



Discrete Time Signal Manipulation

Time shifting

$$x[n] \rightarrow x[n-n_0]$$



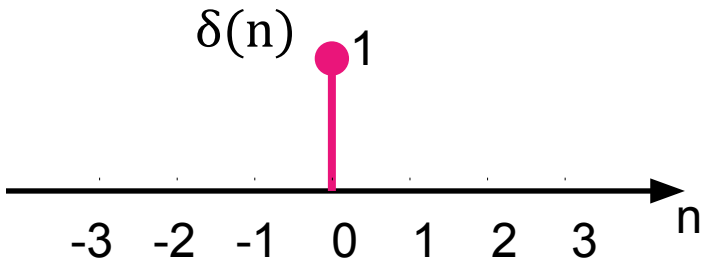
$$w_4[n] = x[n-1]$$

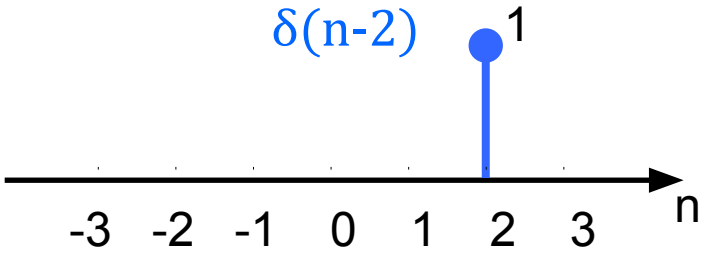
$n_0 > 0 \rightarrow$ time delay

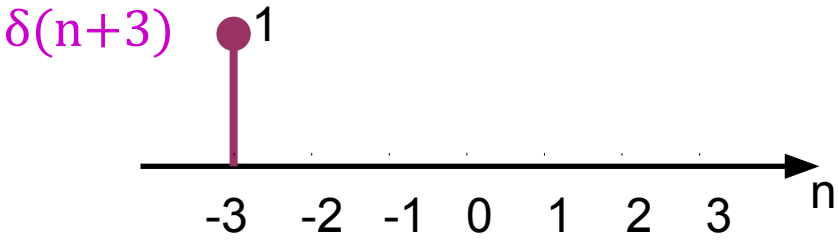
$n_0 < 0 \rightarrow$ time advance

Discrete Time Signal Manipulation

Time shifting

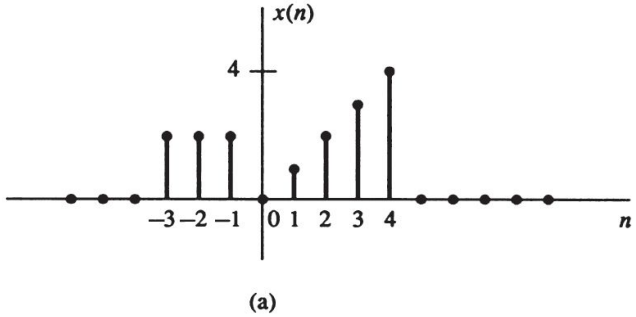




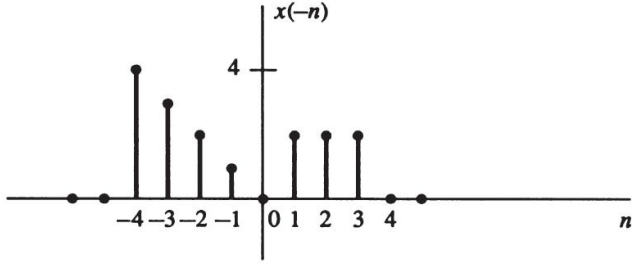


Discrete Time Signal Manipulation

Time scaling & reversal



(a)



(b)

$$x[n] \rightarrow x[kn]$$

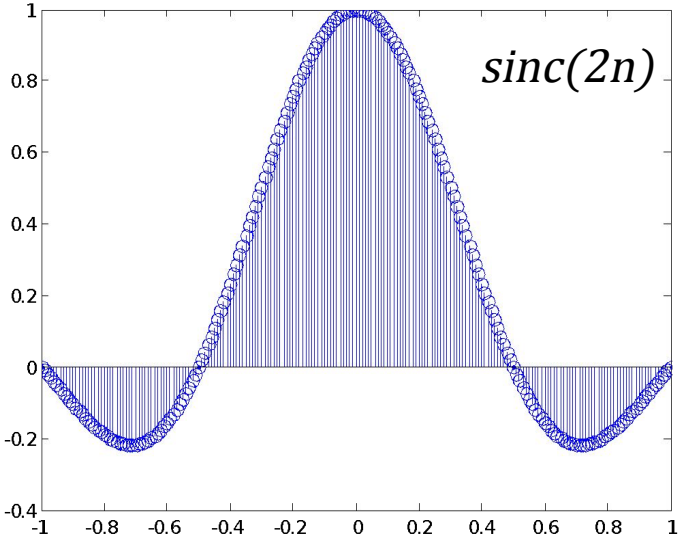
$|k| > 1 \rightarrow$ compress

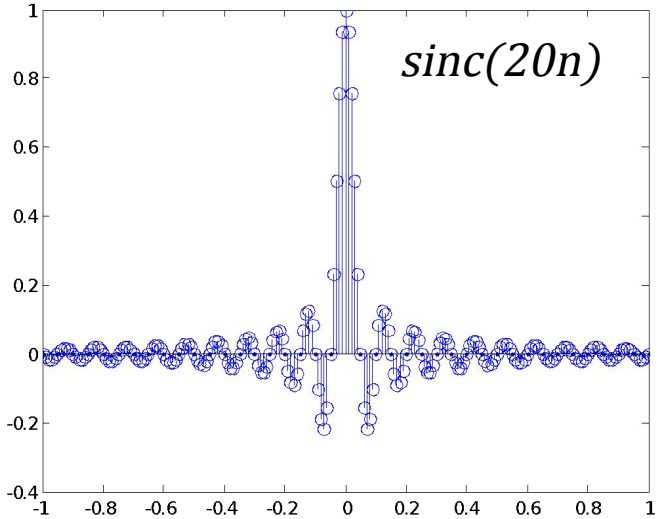
$0 < |k| < 1 \rightarrow$ stretch

$k < 0 \rightarrow$ time reversal/fold

Discrete Time Signal Manipulation

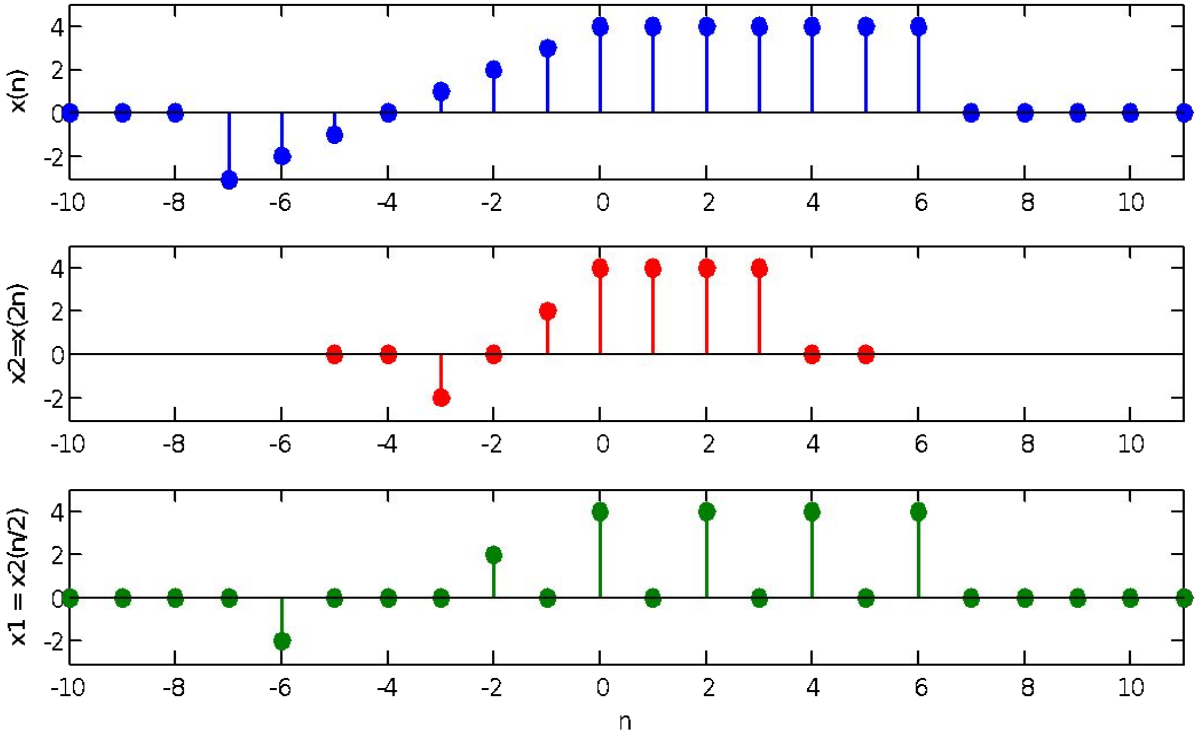
Time scaling





Discrete Time Signal Manipulation

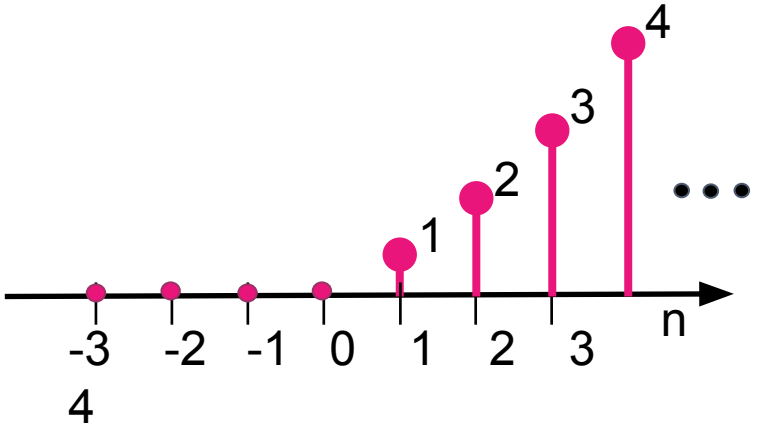
Up-sampling & Down-sampling



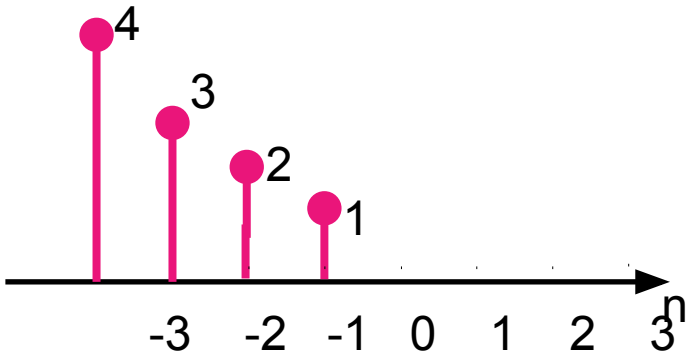
Discrete Time Signal Manipulation

Time reversal

Unit ramp, *r(n)*



Time-reversed unit ramp, *r(-n)*



Notes on Signal Manipulation

- Time Delay and Folding are **NOT** commutative

$$TD[x(n)] = x(n-k), \qquad k > 0$$
$$FD[x(n)] = x(-n)$$

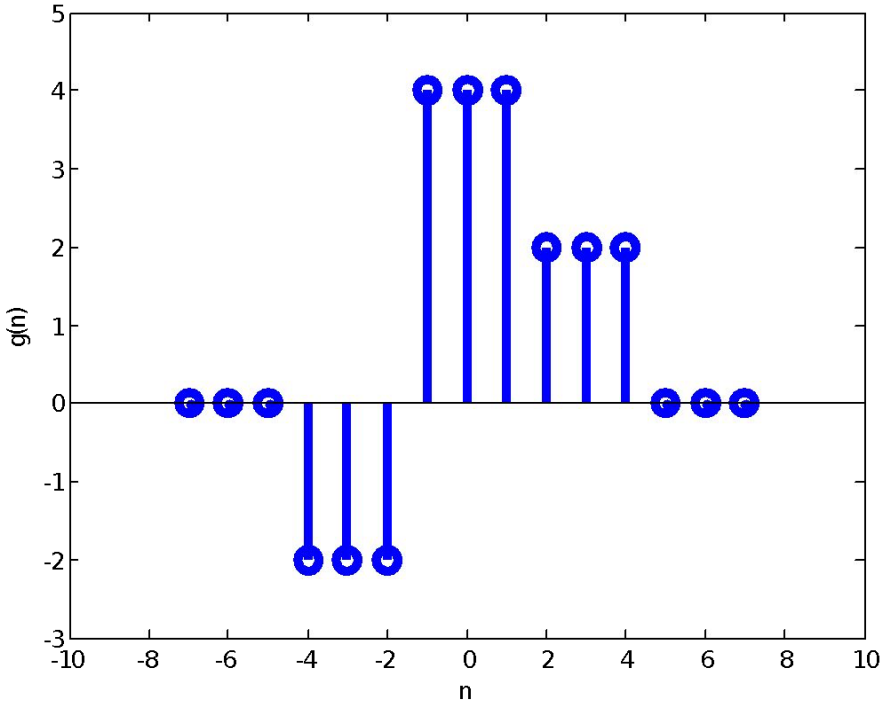
$$TD\{FD[x(n)]\} = TD[x(-n)] = x(-n+k)$$

$$FD\{TD[x(n)]\} = FD[x(n-k)] = x(-n-k)$$

Example:

- Describe the following DTS signal using elementary functions

$$g(n)=\begin{cases}-2, \text{for } -4\le n\le -2\\ 4, \text{for } -1\le n\le 1\\ 2, \text{for } 2\le n\le 4\\ 0, \quad \text{elsewhere}\end{cases}$$



Example:

- **Answer:**

$$-2\delta[n+4] - 2\delta[n+3] - 2\delta[n+2]$$

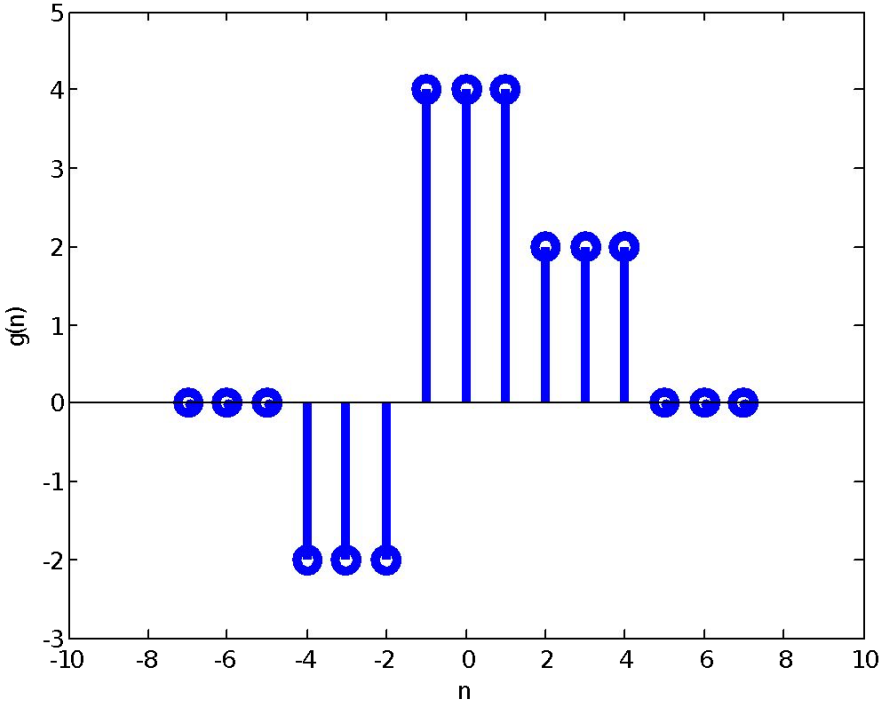
$$+4\delta[n+1] + 4\delta[n] + 4\delta[n-1] +$$

$$2\delta[n-2] + 2\delta[n-3] + 2\delta[n-4]$$

or

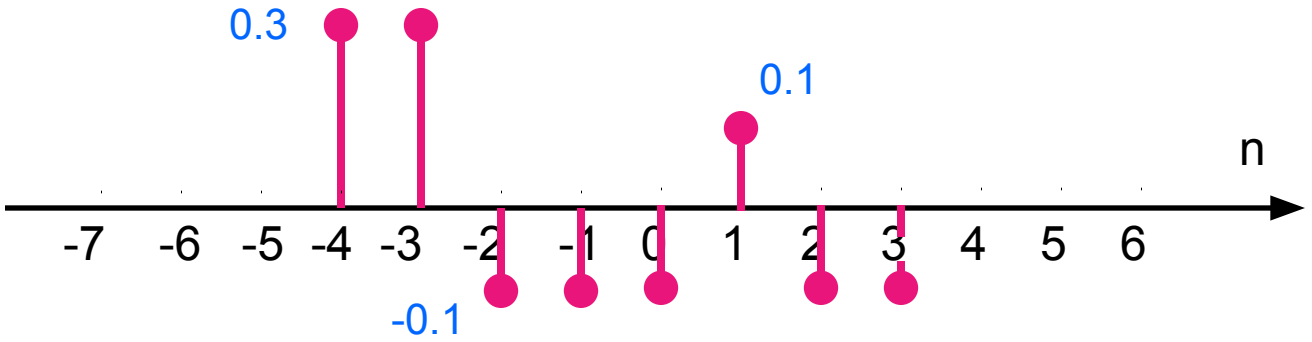
$$-2u[n+4] + 6u[n+1] - 2u[n-2]$$

$$- 2u[n-5]$$



Example:

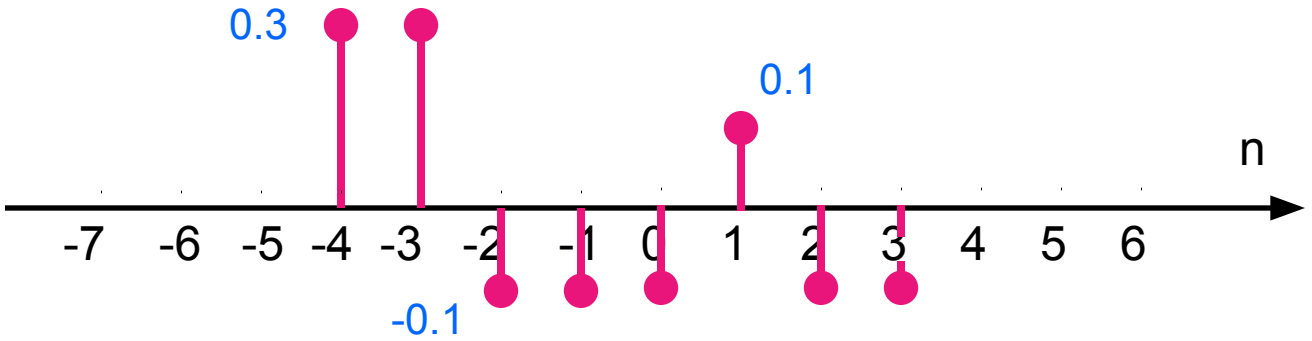
- Describe the following DTS signal using elementary functions



Example:

- **Answer:**

$$x(n) = 0.3u(n+4) - 0.4u(n+2) + 0.2\delta(n-1) + 0.1u(n-4)$$



Even & Odd Signal Decomposition

A discrete time signal can be decomposed into its even and odd components:

$$x_{even}[n] = \frac{x[n]+x[-n]}{2} \qquad x_{odd}[n] = \frac{x[n]-x[-n]}{2}$$

Example :

$x[n] = \{1, \underline{2}, 3, 4, 5\}$

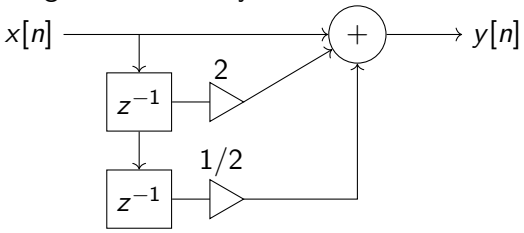
$x_{even}[n] = \{2.5, 2, 2, \underline{2}, 2, 2, 2.5\}$

$x_{odd}[n] = \{-2.5, -2, -1, \underline{0}, 1, 2, 2.5\}$

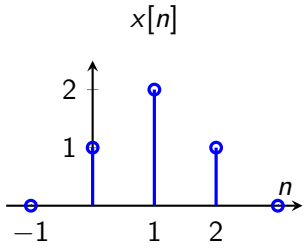
```
>> x = [0 0 1 2 3 4 5];
>> x_fold = fliplr(x)
>> x_even = 0.5*(x+x_fold)
>> x_odd = 0.5*(x-x_fold)
```

Block Diagrams: Iterative solutions

- Given the system block diagram of a MA system:



- Given the input $x[n]$ below, determine the output $y[3]$ and $y[4]$.

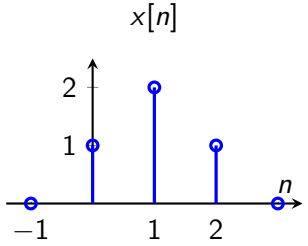


Iterative solution from DE

- The difference equation (DE) describing the system is given by:

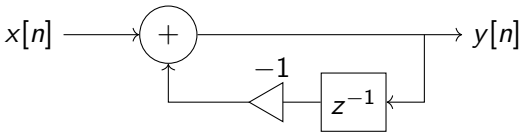
$$y[n] = x[n] + 2x[n-1] + \tfrac{1}{2}x[n-2]$$

- Given the input $x[n]$, determine the output of $y[3]$

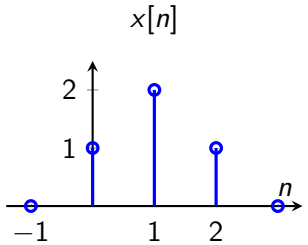


AR System

- Given the system block diagram below with output feedback.

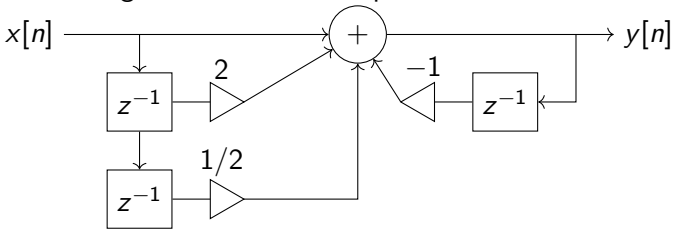


- Given the input $x[n]$ below, determine the output $y[3]$.

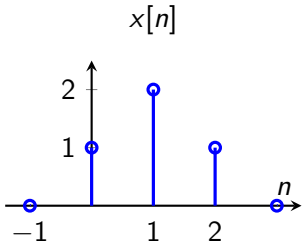


ARMA System

- Given the system block diagram below with output feedback.



- Given the input $x[n]$ below, determine the output $y[3]$ and $y[4]$.



Difference Equations

The **difference equation** is the representation of a system in discrete-time in the same way as a differential equation in continuous-time.

$$y(n) = \alpha y(n-1) + x(n)$$

$$y[k] + \frac{1}{2}y[k-1] + \frac{1}{4}y[k-2] = x[k]$$

$$y[n] = \frac{1}{T}(x[n] - x[n-1])$$

It is a mathematical description of a system as a function of the past, present and future values of the input and output.

Solving the difference equation will allow us to compute for the output for any given input signal.

Difference equations

Difference equations are true for all time n. For example, given the equation:

$$y(n) = \alpha y(n-1) + x(n)$$

If this is true for all n, then its also true if we replace $n = n + 2$. That is

$$y(n+2) = \alpha y(n+1) + x(n+2)$$

The two equations are equivalent.

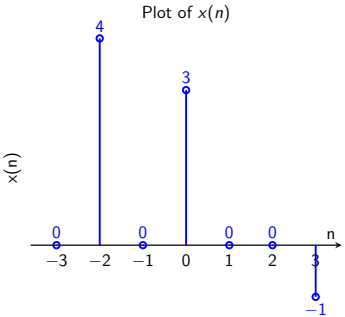
Difference equations

Consider the system described by:

$$y(n) = 5x(n+2) - 3x(n) + 2x(n-1)$$

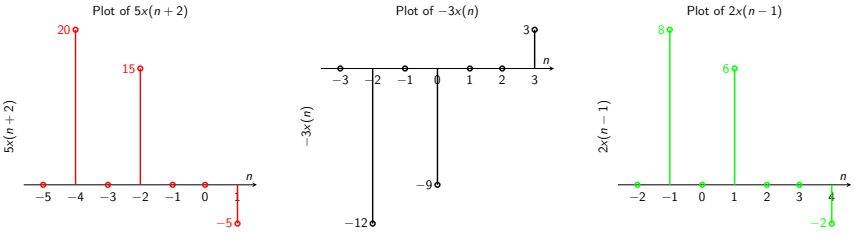
What is the output $y(n)$ when the input is

$$x(n) = 4\delta(n+2) + 3\delta(n) - \delta(n-3)$$



Difference equations

Decomposition of the input signal. Based on the difference equation,the output y(n) can be computed as a linear combination of scaled, shifted versions of the input x(n).



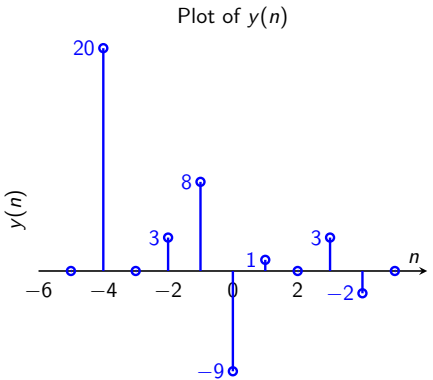
Then add all the plots together to get y(n).

Difference equations

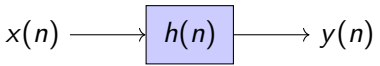
We have

$$y(n) = 20\delta(n+4) + 3\delta(n+2) + 8\delta(n+1) - 9\delta(n) + \delta(n-1) + 3\delta(n-3) - 2\delta(n-4)$$

This method of solving a difference equation becomes tedious as the input becomes more complicated.



Analysis of DT LTI System



In general, the output $y(n)$ of a DT-LTI system with input $x(n)$ and impulse response $h(n)$ can be expressed as a linear combination of past outputs and passed inputs. That is

$$y(n) = \sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k x(n-k)$$

The above difference equation is also called the **input-output equation** and represents one of the ways to describe a **discrete-time linear time-invariant** system (DT-LTI).

Example

Consider the finite-duration sequence given as:

$$x(n) = \{2, \underset{\uparrow}{4}, 0, 3\}$$

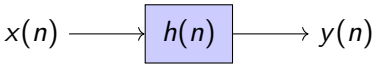
(Note: the $\uparrow$ indicates $n = 0$)

Resolving the sequence as a sum of impulses, we have:

$$x(n) = 2\delta(n+1) + 4\delta(n) + 3\delta(n-2)$$

Convolution Sum

- For an LTI system, we define $h(n)$ as the **unit impulse response**.



  if $x(n) \rightarrow \delta(n)$, then $y(n) \rightarrow h(n)$, that is $h(n) = H[\delta(n)]$
- The output response $y(n)$ is defined as:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

- the above equation is known as the **convolutional sum**
- we say the input is $x(n)$ is convolved with $h(n)$ to yield the output $y(n)$

$$y(n) = x(n) * h(n)$$

Convolution Sum

- The output of the system at an instant say $n = n_0$ is given as:

$$y(n_0) = \sum_{k=-\infty}^{\infty} x(k)h(n_0 - k)$$

That is

$$y(0) = \sum_{k=-\infty}^{\infty} x(k)h(0 - k)$$

$$y(1) = \sum_{k=-\infty}^{\infty} x(k)h(1 - k)$$

And so on.

- The index of summation is k, hence both the input signal $x(k)$ and impulse response $h(k)$ are functions of k.

Convolution Sum

- The output $y(n_0)$ is summed over all values of the product sequence $x(k)h(n_0 - k)$ for all values of k from $-\infty$ to $+\infty$.

- The signal $h(n_0 - k)$ is obtained by "folding" $h(k)$ about $k = 0$ $(h(-k))$, then shifting it by n_0 to obtain $h(n_0 - k)$

About Convolution

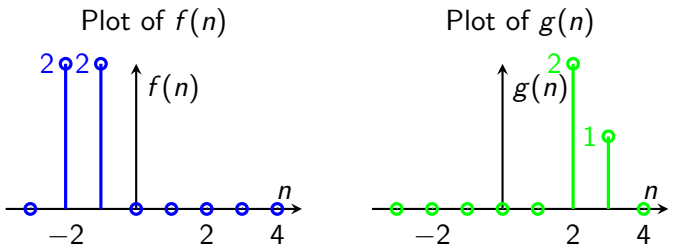
Option You can also choose to fold and shift $x(k) \rightarrow x(-k) \rightarrow x(n-k)$, instead of $h(k)$, this will give the same results since the convolution operation is commutative.

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(n-k)h(k)$$

Interpretation. In the above equation, For each sample of $y(n)$, is obtained as a linear combination of the impulse response $h(k)$, and the coefficients of the linear combination is determined by the values of $x(n-k)$.

Convolution for short signals

Consider the signals $f[n]$ and $g[n]$ as shown. Compute $h[n] = f[n] * g[n]$



Compute the starting point of n: $-2 + 2 = 0$
Compute the end point of n: $(-1) + 3 = 2$
This means that $h[n] = 0$ for $n < 0$ and $n > 2$
Only solve for $h[0]$, $h[1]$ and $h[2]$

Evaluating the convolution

$$h[n] = \sum_{k=-\infty}^{\infty} f[k]g[n-k] = \sum_{k=-2}^{-1} f[k]g[n-k]$$

For $n = 0$:

$$h[0] = \sum_{k=-2}^{-1} f[k]g[0-k] = \sum_{k=-2}^{-1} f[k]g[-k]$$

$$h[0] = f[-2]g[2] + f[-1]g[1] = (2)(2) + (2)(0) = 4$$

For $n = 1$:

$$h[1] = \sum_{k=-2}^{-1} f[k]g[1-k] = f[-2]g[3] + f[-1]g[2] = (2)(1) + (2)(2) = 6$$

For $n = 2$:

$$h[2] = \sum_{k=-2}^{-1} f[k]g[2-k] = f[-2]g[4] + f[-1]g[3] = (2)(0) + (2)(1) = 2$$

Useful Table of Convolution Operations

Table: Useful convolution sums

$\mathbf{x_1[n]}$	$\mathbf{x_2[n]}$	$\mathbf{x_1[n] * x_2[n]}$
$\delta[n-k]$	$x[n]$	$x[n-k]$
$u[n]$	$u[n]$	$(n+1)u[n]$
$\alpha^n u[n]$	$u[n]$	$\left[\frac{1-\alpha^{n+1}}{1-\alpha}\right]u[n]$
$\alpha^n u[n]$	$\alpha^n u[n]$	$(n+1)\alpha^n u[n]$
$\alpha^n u[n]$	$\beta^n u[n]$	$\left[\frac{\alpha^{n+1}-\beta^{n+1}}{\alpha-\beta}\right]u[n]$

Iterative Solution

Given the 2nd-order difference equation:

$$y[n+2] - y[n+1] + 0.3y[n] = x[n+2] - 3x[n+1]$$

with the initial conditions $y[-1] = 1$ and $y[-2] = 2$ and $x[n] = nu[n]$. Rewriting the equation as:

$$y[n+2] = y[n+1] - 0.3y[n] + x[n+2] - 3x[n+1]$$

Case $n = -2$: $y[0] = y[-1] - 0.3y[-2] + x[0] - 3x[-1]$

$$y[0] = 1 - (0.3)(2) + 0 - 0 = 1 - 0.6 = 0.4$$

Case $n = -1$: $y[1] = y[0] - 0.3y[-1] + x[1] - 3x[0]$

$$y[1] = 0.4 - (0.3)(1) + 1 - 0 = 1.1$$

Continue solving the equation for all value of n iteratively.

Zero-input response and Characteristic Polynomial

Zero-input response: This is the response of the system when $x[n] = 0$, due to the initial conditions. (homogeneous solution). That is in the general form of a DT-LTI system, we have:

$$y(n) = \sum_{k=1}^{N} a_k y(n-k)$$

The solution to this equation is of the form λ^n. We have

$$\sum_{k=0}^{N} a_k \lambda^{n-k} = 0$$

Expanding

$$a_0\lambda^n + a_1\lambda^{n-1} + ... + a_{N-1}\lambda^{n-N+1} + a_N\lambda^{n-N} = 0$$

Factoring λ^{n-N} :

$$\lambda^{n-N} \underbrace{(a_0\lambda^N + a_1\lambda^{N-1} + ... + a_{N-1}\lambda + a_N)}_{\text{characteristic polynomial}} = 0$$

Zero-input Response

The **characteristic polynomial** is of the form

$$(a_0\lambda^N + a_1\lambda^{N-1} + ... + a_{N-1}\lambda + a_N)$$

In general, it has N roots $(\lambda_1, \lambda_2, ..., \lambda_N)$, which can be real and distinct, repeated roots or complex conjugates, since the coefficients $a_0, a_1, ..., a_N$ are real.

Zero-input Response: The zero-input response is a linear combination of $C_i\lambda^n$ (characteristic modes) and is the general solution to the equation.

$$y_{zi}(n) = C_1\lambda_1^n + C_2\lambda_2^n + ... + C_N\lambda_N^n$$

whee $C_1, C_2, ..., C_N$ are constants, and they are solved by applying the initial conditions of the system.

Solving for the Zero-Input response

Consider the DT-LTI system described by

$$y[n+2] - 0.8y[n+1] - 0.09y[n] = 3x[n+2]$$

with initial conditions $y[-1] = 0, y[-2] = \frac{10}{3}$.

The characteristic equation is

$$\lambda^2 - 0.8\lambda - 0.09 = 0$$

$$(\lambda - 0.9)(\lambda + 0.1) = 0$$

Therefore; The roots (eigenvalues) of the characteristic equation are $\lambda_1 = 0.9$ and $\lambda_2 = -0.1$

Solving for the Zero-Input response

The roots are real and distinct. The zero-input response can be written as:

$$y_{zi}(n) = C_1(0.9)^n + C_2(-0.1)^n$$

Solving for C_1 and C_2 using the initial conditions.

Let $n = -1$: $y[-1] = 0 = C_1(0.9)^{-1} + C_2(-0.1)^{-1}$

$$C_1\tfrac{10}{9} + C_2(10) = 0$$

Let $n = -2$: $y[-2] = \frac{10}{3} = C_1(0.9)^{-2} + C_2(-0.1)^{-2}$

$$C_1\tfrac{100}{81} + C_2(100) = \tfrac{10}{3}$$

Solving, $C_1 = \frac{27}{100}$ and $C_2 = \frac{3}{100}$, the **zero-input response** is

$$y_{zi}(n) = \tfrac{27}{100}(0.9)^n + \tfrac{3}{100}(-0.1)^n$$

Example: Zero-Input Response

For a DT-LTI system described by the difference equation, determine the zero-input response.

$$y[n+2] - 0.6y[n+1] - 0.16 = 5x[n+2]$$

given the initial conditions $y[-1] = 0$ and $y[-2] = \frac{25}{4}$.

The characteristic equation is

$$\lambda^2 - 0.6\lambda - 0.16 = (\lambda + 0.2)(\lambda + 0.8) = 0$$

The roots are $\lambda_1 = -0.2$ and $\lambda_2 = 0.8$
The zero-input response is:

$$y_{zi}[n] = C_1(-0.2)^n + C_2(0.8)^n$$

Example: Zero-Input Response

Solving for the constants.

Let $n = -1$: $0 = -5C_1 + \frac{5}{4}C_2$

Let $n = -2$: $\frac{25}{4} = 25C_1 + \frac{25}{16}C_2$

The constants are $C_1 = \frac{1}{5}$ and $C_2 = \frac{4}{5}$

The **zero-input reponse** is:

$$y_{zi}[n] = \frac{1}{5}(-0.2)^n + \frac{4}{5}(0.8)^n$$

Example: Repeated roots

Given the DT-LTI system described by:

$$y[n+2] + 6y[n+1] + 9y[n] = 2x[n+2] + 6x[n]$$

with initial conditions: $y[-1] = -\frac{2}{3}$ and $y[-2] = \frac{1}{9}$

The characteristic equation is $\lambda^2 + 6\lambda + 9 = 0$

$$(\lambda + 3)^2 = 0$$

The roots are: $\lambda_1 = -3$ and $\lambda_2 = -3$

Example: Repeated roots

The roots are repeated. The zero-input response is written as:

$$y_{zi}(n) = C_1(-3)^n + C_2 n(-3)^n$$

Solving for C_1 and C_2 from the initial conditions.

Let $n = -1$: $-\frac{2}{3} = C_1(-3)^{-1} + C_2 n(-3)^{-1}$

$$-\frac{2}{3} = (C_1 - C_2)(-\tfrac{1}{3})$$

Let $n = -2$: $\frac{1}{9} = C_1(-3)^{-2} + C_2 n(-3)^{-2}$

$$\tfrac{1}{9} = (C_1 - 2C_2)(\tfrac{1}{9})$$

Solving: $C_1 = 3$ and $C_2 = 1$, the zero-input response is

$$y_{zi}(n) = 3(-3)^n + n(-3)^n = (-3)^n(3 + n)$$

Example: Complex roots

Given the DT-LTI system described by:

$$y[n+2] - 1.3y[n+1] + 0.663y[n] = x[n+2]$$

with initial conditions $y[-1] = 2$ and $y[-2] = 1$.

The characteristic equation is $\lambda^2 - 1.3\lambda + 0.663 = 0$

$$(\lambda - 0.65 + j0.49)(\lambda - 0.65 - j0.49) = 0$$

The roots are $\lambda = 0.65 \pm j0.49$, in polar form $\lambda \cong 0.814e^{\pm j0.646}$

The form of the zero-input response is

$$y_{zi}(n) = C|\lambda|^n cos(\beta n + \theta)$$

where C and θ are unknown parameters that are solved using the initial conditions

Example: Complex roots

The zero-input response is

$$y_{zi}(n) = C|\lambda|^n cos(\beta n + \theta)$$

Solving the parameters using the initial conditions.

$$|\lambda| = |0.819e^{+j0.646j}| = |0.814| \underbrace{|e^{+j0.646}|}_{\text{magnitude}=1} = 0.814$$

$$\beta = \measuredangle 0.814e^{j0.646} = 0.646$$

So we have:

$$y_{zi}(n) = C|0.814|^n cos(0.646n + \theta)$$

Using the two initial conditions, solve for the two unknowns C and θ

Example: Complex roots

Given

$$y_{zi}(n) = C|0.814|^n cos(0.646n + \theta)$$

Using the two initial conditions, solve for the two unknowns C and θ.

Let $n = -1$: $2 = \frac{C}{0.814} cos(0.646(-1) + \theta)$

Let $n = -2$: $1 = \frac{C}{0.663} cos(0.646(-2) + \theta)$

Solving for C and θ (see details in the Appendix) we have:

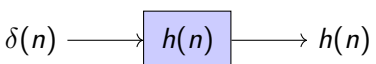
$$C = 1.942 \text{ and } \theta = 0.0696 \text{ radians}$$

The **zero-input response** is

$$y_{zi}(n) = 1.942|0.814|^n cos(0.646n + 0.0696)$$

Impulse response

The impulse response $h(n)$ of a DT-LTI system is the output response, with an impulse $\delta(n)$ as input and the system is initially at rest, or with **zero-initial states**. (zero initial conditions).



We can use the **iterative method** to solve for the first few samples of the **impulse response** $h(n)$

Impulse response

Determine the **impulse response** of the DT systems described by:

$$y[n] - 0.6y[n-1] - 0.16y[n-2] = 5x[n]$$

It follows that $x[n] = \delta[n]$ and $y[n] = h[n]$

$$h[n] - 0.6h[n-1] - 0.16h[n-2] = 5\delta[n]$$

Initial conditions: system is initially at rest, so

$$h[-1], h[-2], h[-3], etc = 0$$

We rewrite the equation for iterative solution as:

$$h[n] = 0.6h[n-1] + 0.16h[n-2] + 5\delta[n]$$

Impulse response: Iterative solution

At $n = 0$: $h[0] = 0.6\underbrace{h[-1]}_{0} + 0.16\underbrace{h[-2]}_{0} + 5\underbrace{\delta[0]}_{1} = 5$

At $n = 1$: $h[1] = 0.6\underbrace{h[0]}_{5} + 0.16\underbrace{h[-1]}_{0} + 5\underbrace{\delta[1]}_{0} = (0.6)(5) = 3$

At $n = 2$: $h[2] = 0.6h[1] + 0.16h[0] + 5\delta[2] = (0.6)(3) + (0.16)(5) = 2.6$

Continue solving for $h[3], h[4], h[5]$, etc. iteratively.

Being able to solve for a few values of $h(n)$ as initial conditions, can help solve for the closed-form solution of the impulse response $h(n)$

System response to external input: Zero-state response

The **zero-state** response $y[n]$ is the system response to an input $x[n]$, when the system is in zero-state, or zero-initial conditions.

The **zero-state response** $y_{zs}[n]$ of the system is the input $x[n]$ convolved with the impulse response $h[n]$.

$$y_{zs}[n] = x[n] * h[n]$$

Zero-state response

Calculate the **zero-state** response $y_{zs}[n]$ of a DT-LTI system described by:

$$y[n+2] - 0.6y[n+1] - 0.16y[n] = 5x[n+2]$$

given the input $x[n] = (4^{-n})u[n]$

In the previous example, the **impulse response** of this system is:

$$h[n] = [(-0.2)^n + 4(0.8)^n]u[n]$$

The zero-state response is $y_{zs}[n] = x[n] * h[n]$

$$y_{zs}[n] = (4^{-n})u[n] * [(-0.2)^n + 4(0.8)^n]u[n]$$

$$y_{zs}[n] = (4^{-n})u[n] * (-0.2)^n u[n] + (4^{-n})u[n] * 4(0.8)^n u[n]$$

Zero-state Response

Note that $(4^{-n})u[n] = (\frac{1}{4})^n u[n] = (0.25)^n u[n]$
Therefore

$$y_{zs}[n] = (0.25^n)u[n] * (-0.2)^n u[n] + 4(0.25^n u[n] * (0.8)^n u[n]$$

From the table of convolutional sum:

$$\begin{aligned} y_{zs}[n] &= \left[\frac{(0.25)^{n+1} - (-0.2)^{n+1}}{0.25 - (-0.2)} + 4\frac{(0.25)^{n+1} - (0.8)^{n+1}}{0.25 - 0.8}\right] u[n] \\ &= \left(2.22[(0.25)^{n+1} - (-0.2)^{n+1}] - 7.27[(0.25)^{n+1} - (0.8)^{n+1}]\right) u[n] \\ &= [-5.05(0.25)^{n+1} - 2.22(-0.2)^{n+1} + 7.27(0.8)^{n+1}]u[n] \end{aligned}$$

The **zero-state response** of the system is:

$$y_{zs}[n] = [-1.26(0.25)^n + 0.444(-0.2)^n + 5.81(0.8)^n]u[n]$$

Total Response

The **total response** of a DT-LTI system can be expressed as the sum of the zero-input response and the zero-state response:

$$y_{total}[n] = y_{zi}[n] + y_{zs}[n]$$

where

$$y_{total}[n] = \underbrace{\sum_{j=1}^{n} C_j \lambda_j^n}_{\text{zero-input}} + \underbrace{x[n] * h[n]}_{\text{zero-state}}$$

Example: Total Response

Determine the **total response** of the DT-LTI system defined by

$$y[n+2] - 0.16y[n+1] - 0.16y[n] = 5x[n+2]$$

with initial conditions $y[-1] = 0$, $y[-2] = \frac{25}{4}$ and input $x[n] = (4)^{-n}u[n]$ From the previous examples we have determined that the **impulse response** is given as:

$$h[n] = [(-0.2)^n + 4(0.8)^n]u[n]$$

The **zero-input response** as:

$$y_{zi}[n] = \frac{1}{5}(-0.2)^n + \frac{4}{5}(0.8)^n$$

And the **zero-state response** is:

$$y_{zs}[n] = [-1.26(0.25)^n + 0.444(-0.2)^n + 5.81(0.8)^n]u[n]$$

Example: Total Response

The total response therefore is given as:

$$y_t[n] = \underbrace{(0.2(-0.2)^n + 0.8(0.8)^n}_{\text{zero-input component}}$$

$$\underbrace{-1.26(0.25)^n + 0.444(-0.2)^n + 5.81(0.8)^n)}_{\text{zero-state component}} u[n]$$

Note: The characteristic modes of this system are $(-0.2)^n$ and $(0.8)^n$. The zero-input component is made up of characteristic modes exclusively, but there are also characteristic modes in the zero-state response. When all the characteristic modes are lumped together, the resulting component is the **natural response** and the remaining part is the **forced response**. In this example:

$$y_t[n] = \underbrace{0.644(-0.2)^n + 6.61(0.8)^n}_{\text{Natural response}} \underbrace{-1.26(4)^{-n}}_{\text{Forced response}} u[n]$$

Impulse response: Closed-form solution

The impulse response of the system cam be computed from the zero-input response, with the initial conditions dictated by the impulse $\delta(n)$ at $n = 0$. That is for a causal $h(n)$

$$h(n) = \frac{b_0}{a_0}\delta(n) + y_{zi}(n)u(n)$$

where the coefficients a_0 and b_0 are the coefficients of the general-form of the difference equation, and $y_{zi}(n)$ is a linear combination of the characteristic modes of the system.

The n unknown coefficients in $y_{zi}(n)$ can be determined from knowledge of the first n values of $h(n)$. Fortunately, this can be computed iteratively as demonstrated earlier.

Impulse response: Closed-form Solution

Determine the impulse response $h(n)$ specified by the difference equation:

$$y[n] - 0.6y[n-1] - 0.16y[n-2] = 5x[n]$$

The equation can be advanced by setting $n = n + 2$

$$y[n+2] - 0.6y[n+1] - 0.16y[n] = 5x[n+2]$$

Therefore $a_0 = -0.16$ and $b_0 = 0$.
The characteristic equation is:

$$\lambda^2 - 0.6\lambda - 0.16 = (\lambda + 0.2)(\lambda - 0.8)$$

The characteristic modes are: $(-0.2)^n$ and $(0.8)^n$
The **zero-input response** is:

$$y_{zi}[n] = C_1(-0.2)^n + C_2(0.8)^n$$

Impulse response: Closed-form solution

The **impulse response** therefore is:

$$h[n] = (C_1(-0.2)^n + C_2(0.8)^n)u[n]$$

To determine the values of C_1 and C_2, we need two values of $h[n]$. This has been computed earlier with $h[0] = 5$ and $h[1] = 3$.

Using the initial conditions, we have $5 = C_1 + C_2$ and $3 = 0.2C_1 + 0.8C_2$. Therefore,

$$C_1 = 1 \text{ and } C_2 = 4$$

The **impulse response** is:

$$h[n] = [(-0.2)^n + 4(0.8)^n]u[n]$$

Appendix: Complex roots complete solution

Solving for the unknown parameters C and θ given the two equations and applying the initial conditions.

Let $n = -1$: $2 = \frac{C}{0.814}\cos(0.646(-1) + \theta)$

Let $n = -2$: $1 = \frac{C}{0.663}\cos(0.646(-2) + \theta)$

Using the trigonometric identity $\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$

$$2 = \frac{C}{0.814}[\cos(0.646)\cos(\theta) + \sin(0.646)\sin(\theta)]$$

$$1 = \frac{C}{0.663}[\cos(1.292)\cos(\theta) + \sin(1.292)\sin(\theta)]$$

Complex roots complete solution

Simplifying we have:

$$2 = C[0.981\cos(\theta) + 0.740\sin(\theta)] = 0.981C\cos(\theta) + 0.740C\sin(\theta)$$

$$1 = C[0.415\cos(\theta) + 1.450\sin(\theta)] = 0.415\underbrace{C\cos(\theta)}_{x} + 1.450\underbrace{C\sin(\theta)}_{y}$$

We have

$$0.981x + 0.740y = 2 \text{ and } 0.415x + 1.450y = 1$$

Solving for x and y: $x = 1.937 = C\cos(\theta)$ and $y = 0.135 = C\sin(\theta)$

$$\frac{C\sin(\theta)}{C\cos(\theta)} = \tan(\theta) = \frac{0.135}{1.937} \Rightarrow \theta = \tan^{-1}(\tfrac{0.135}{1.937})$$

Therefore: $\theta = 0.0696$ radians and $C = 1.942$

Appendix: Classical Method

We present the classical method of solving difference equations. The total solution $y_t[n]$ is the sum of the homogeneous solution and the particular solution.

$$y_t[n] = y_h[n] + y_p[n]$$

where $y_h[n]$: **homogeneous solution** is the solution when the input is zero. i.e. zero-input response
and $y_p[n]$: **particular solution** takes the form of the input.

Particular solution

The particular solution is assumed to take the form of the input, the table shows the general forms of $y_p[n]$ given $x[n]$

Table: General forms of $y_p[n]$

$x[n]$	$y_p[n]$
A (constant)	K (constant)
$A(m)^n$ (m is constant)	$K(M)^n$ (M is constant)
An^M	$K_0n^M + K_1n^{M-1} + ... + K_M$
A^nn^M	$A^n(K_0n^M + K_1n^{M-1} + ... + K_M)$
$Acos(\omega_0 n)/Asin(\omega_0 n)$	$K_1cos(\omega_0 n) + K_2sin(\omega_0 n)$

Example: Particular solution

Determine the particular solution of

$$y[n] = \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] + x[n]$$

when $x[n] = 2^n, n \geq 2$ and 0 elsewhere

The form of the particular solution is $y_p[n] = K(2)^n, n \geq 0$ according to the table. Since its a solution it should satisfy the difference equation, substituting:

$$K(2)^n u[n] = \frac{5}{6}K2^{n-1}u[n-1] - \frac{1}{6}K2^{n-2}u[n-2] + 2^n u[n]$$

To solve for K, find a value of n where none of the terms vanish, for this case $n = 2$, since all terms exist for $n \geq 2$.

Particular solution

Since $u[2] = u[1] = u[0] = 1$, we have

$$4K = \frac{5}{6}K(2) - \frac{1}{6}K + 4$$

Solving for K, we have $K = \frac{8}{5}$

Therefore

$$y_p[n] = \frac{8}{5}(2)^n u[n]$$

Total solution

Solve for the **total solution** of the system described by

$$y[n] = -0.5y[n-1] + 0.5y[n-2] + x[n] - x[n-1]$$

with input $x[n] = (0.5)^n u[n]$, initial conditions $y[-1] = 1$, $y[-2] = -1$

Homogeneous solution $y_h[n]$:

$$y_h[n] + \tfrac{1}{2}y_h[n-1] - \tfrac{1}{2}y_h[n-2] = 0$$

The characteristic equations is:

$$\lambda^2 + \tfrac{1}{2}\lambda - \tfrac{1}{2} = (\lambda + 1)(\lambda - \tfrac{1}{2}) \rightarrow \lambda_1 = -1, \lambda_2 = \tfrac{1}{2}$$

The **homogeneous solution** is:

$$y_h[n] = C_1(-1)^n + C_2(\tfrac{1}{2})^n$$

Particular solution

Since the input is $x[n] = (0.5)^n u[n]$, has the form that is equal to one of the homogeneous solutions.
The form of the **particular solution** is:

$$y_p[n] = Kn(0.5)^n u[n]$$

Since $y_p[n]$ is a solution, we can substitute:

$$Kn(\tfrac{1}{2})^n u[n] = -\tfrac{1}{2}K(n-1)(\tfrac{1}{2})^{n-1}u[n-1] + \tfrac{1}{2}K(n-2)(\tfrac{1}{2})^{n-2}u[n-2] + (\tfrac{1}{2})^n u[n] - (\tfrac{1}{2})^{n-1}u[n-1]$$

Solving for K, at $n = 2$ none of the terms vanish, we have:

$$K2(\tfrac{1}{2})^2 = -\tfrac{1}{2}K(2-1)(\tfrac{1}{2})^{2-1} + (\tfrac{1}{2})K(2-2)(\tfrac{1}{2})^{2-2} + (\tfrac{1}{2})^2 - (\tfrac{1}{2})^{2-1}$$

$$\tfrac{1}{2}K = -\tfrac{1}{4}K + \tfrac{1}{4} - \tfrac{1}{2} \rightarrow K = -\tfrac{1}{3}$$

Total solution

The total solution therefore is:

$$y_t[n] = [C_1(-1)^n + C_2(\tfrac{1}{2})^n - \tfrac{1}{3}n(\tfrac{1}{2})^n]u[n]$$

To solve for C_1 and C_2, we have to use $n \geq 0$, so that none of the terms vanish. We cannot use $y[-1]$ and $y[-2]$.

We can solve for $y[0]$ and $y[1]$ by iteration from the original difference equation:

$$y[n] = -0.5y[n-1] + 0.5y[n-2] + x[n] - x[n-1]$$

with input $x[n] = (0.5)^n u[n]$, initial conditions $y[-1] = 1$, $y[-2] = -1$

Total solution

$$\text{At } n = 0 : y[0] = -\tfrac{1}{2}y[-1] + \tfrac{1}{2}y[-2] + x[0] - x[-1]$$
$$= -\tfrac{1}{2}(1) + \tfrac{1}{2}(-1) + 1 = 0$$

$$\text{At } n = 1 : y[1] = -\tfrac{1}{2}y[0] + \tfrac{1}{2}y[-1] + x[1] - x[0]$$
$$= -\tfrac{1}{2}(0) + \tfrac{1}{2}(1) + \tfrac{1}{2} - 1 = 0$$

Using $y[0] = 0$ and $y[1] = 0$ as initial conditions, and using the expression for $y_t[n]$ we have:

$$y[0] = C_1 + C_2 = 0$$

$$y[1] = -C_1 + \tfrac{1}{2}C_2 - \tfrac{1}{6} = 0$$

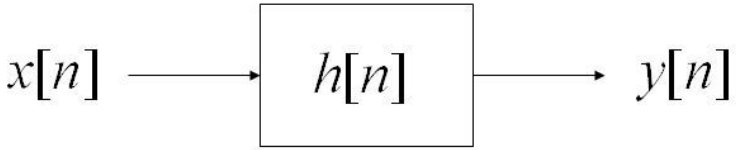
Solving we have: $C_1 = -\frac{1}{9}$ and $C_2 = \frac{1}{9}$, the **total solution** is

$$y_t[n] = [-\tfrac{1}{9}(-1)^n + \tfrac{1}{9}(\tfrac{1}{2})^n - \tfrac{1}{3}n(\tfrac{1}{2})^n]u[n]$$

DTS Convolution

Convolutional Sum of 2 DTS, x[n] and h[n]:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

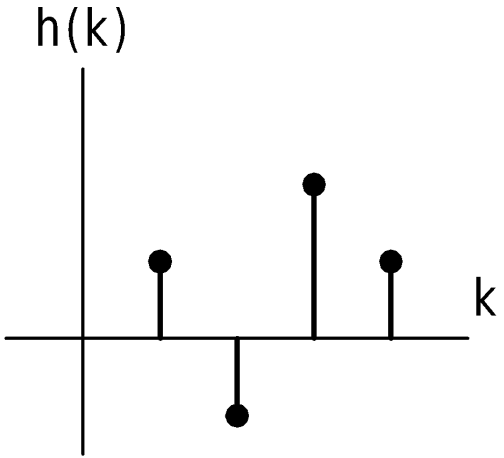


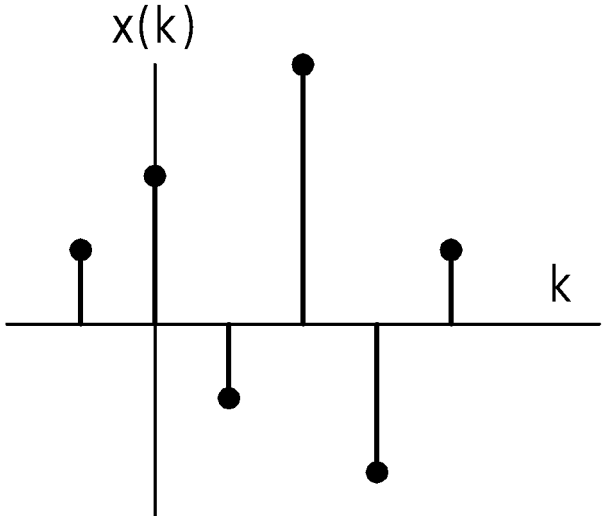
- Denoted as **y[n] = x[n]*h[n]** or **y[n] = h[n]*x[n]**
- Involves 1) Folding, 2)Shifting, 3) Multiplication, and 4) Addition
- Convolution is commutative as well as associative

Example:

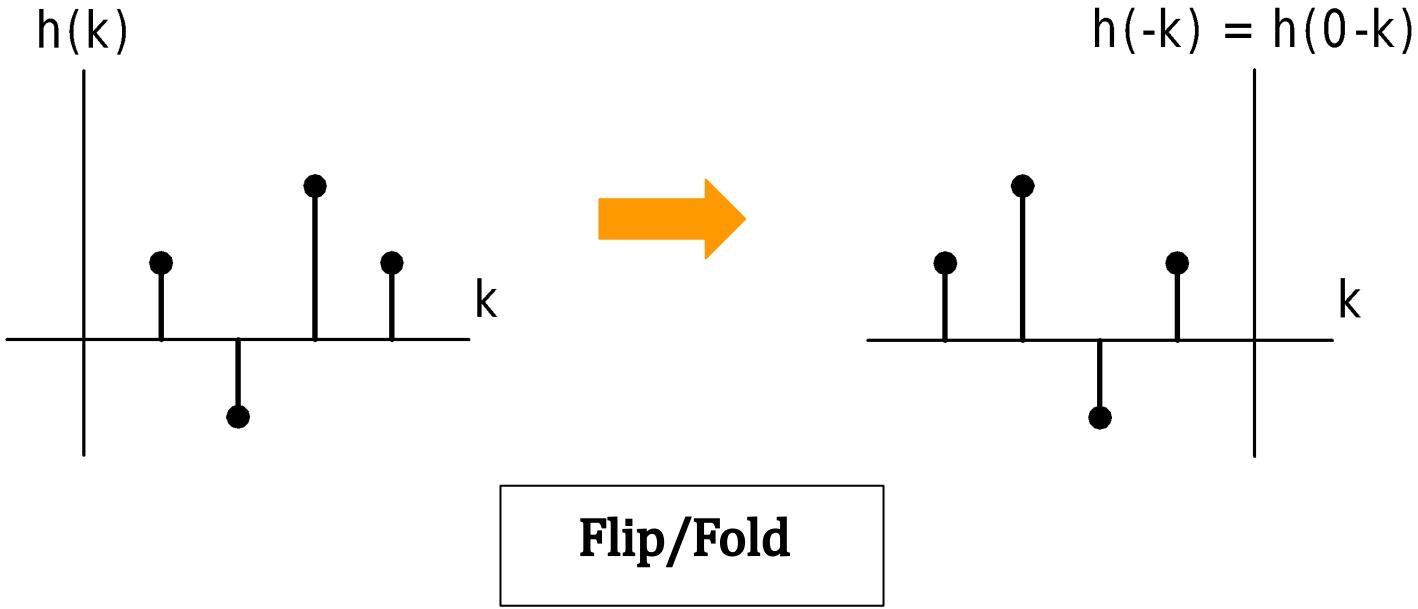
Convolve the following signals:

$h[n] = \{\underline{0}, 1, -1, 2, 1\}$ and $x[n] = \{1, \underline{2}, -1, 3, -2, 1\}$

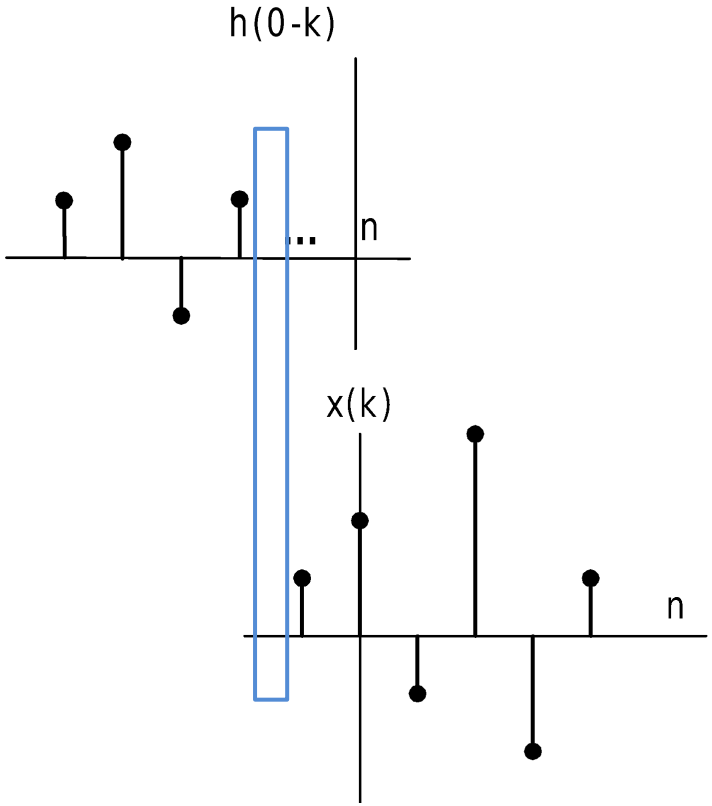




DTS Convolution

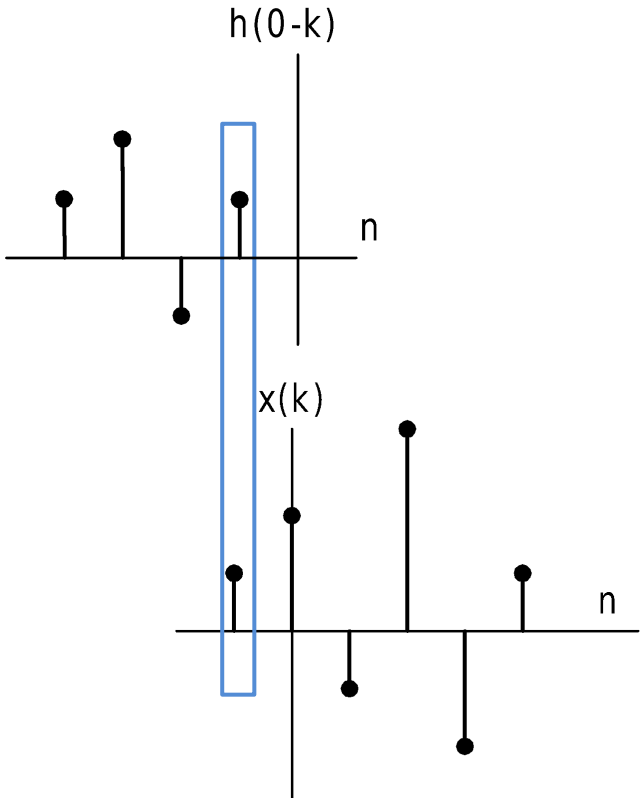


DTS Convolution



Shift:
Note that when $n_0 < 0$,
there are no overlapping signals: $y(-1), y(-2), \ldots = 0$

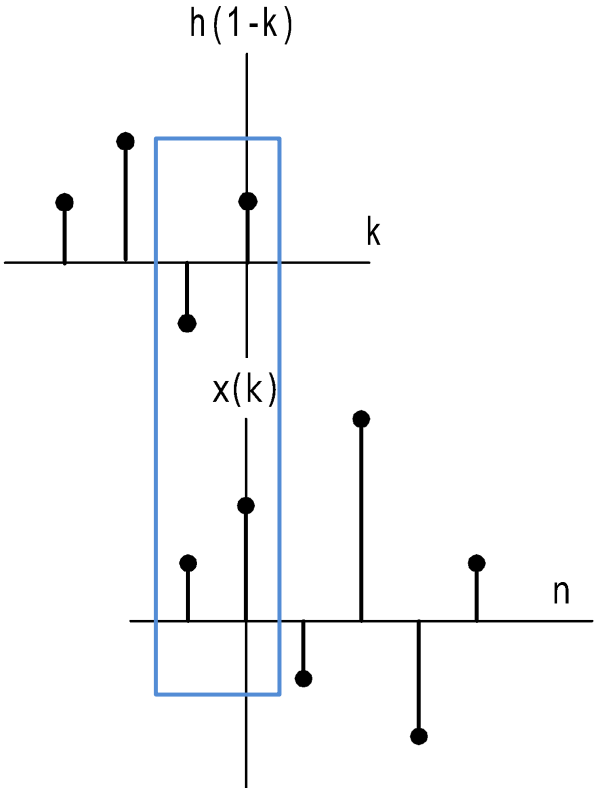
DTS Convolution



Shift:
Note that when $n_0 = 0$,

$$y(0) = 1$$

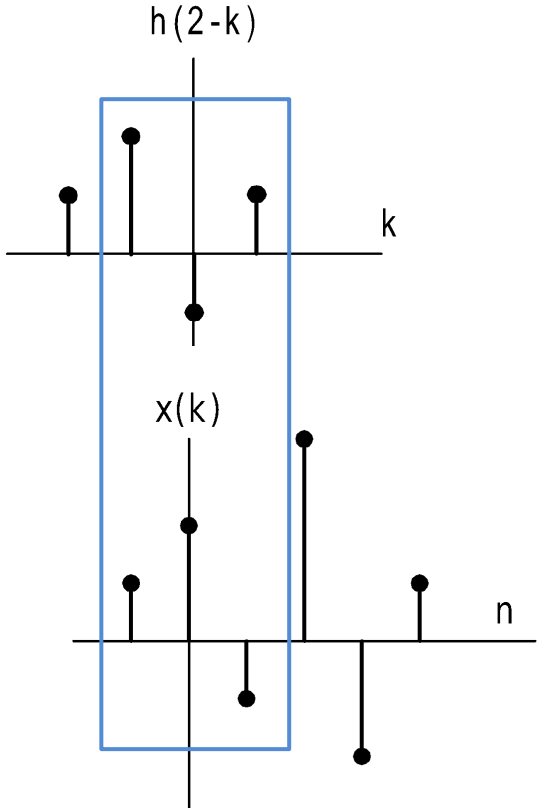
DTS Convolution



Shift:
Note that when $n_0 = 1$,

$$y(1) = -1(1) + 1(2) = 1$$

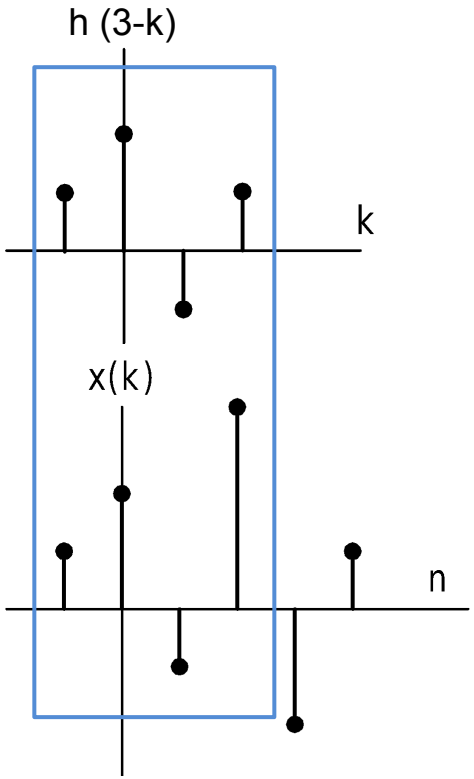
DTS Convolution



Shift:
Note that when $n_0 = 2$,

$$y(2) = 2(1) - 1(2) + 1(-1) = -1$$

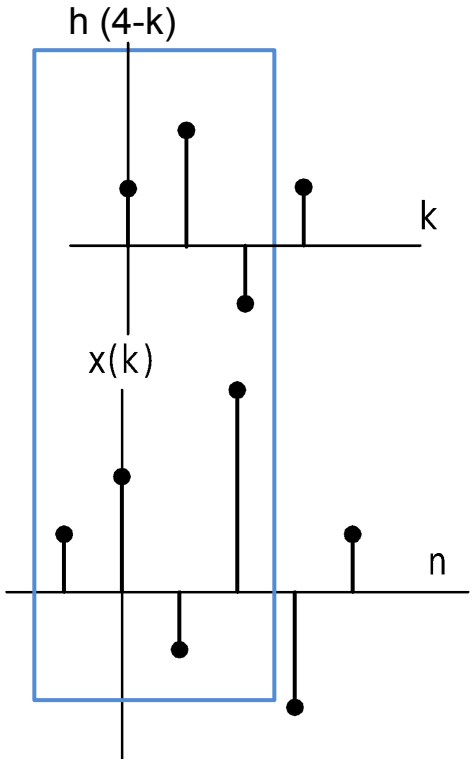
DTS Convolution



Shift:
Note that when $n_0 = 3$,

$$y(3) = 1(1) + 2(2) + -1(-1) + 1(3) = 9$$

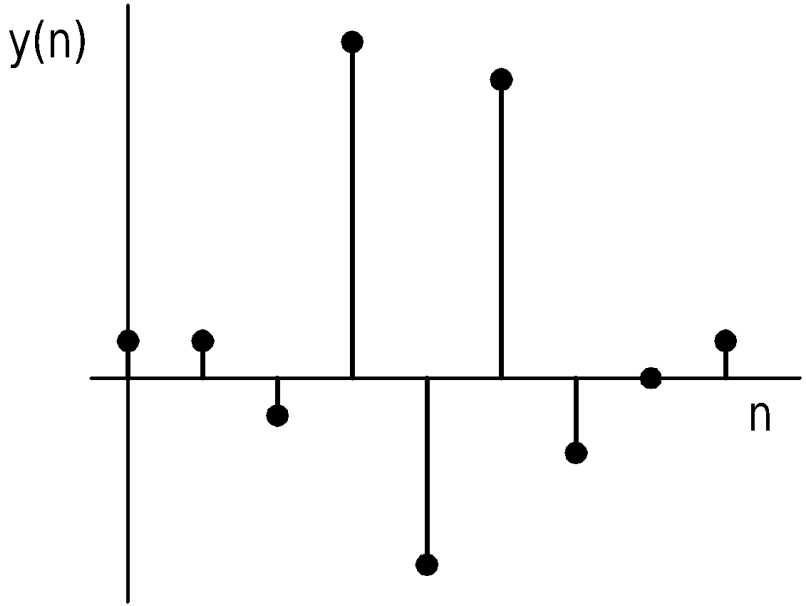
DTS Convolution



Shift:
Note that when $n_0 = 4$,

$y(4) = 1(2) + 2(-1) + -1(3) + 1(-2) = -5$

DTS Convolution



$y(n) = \{ \underline{1}, 1, -1, 9, -5, 8, -2, 0, 1 \}$

```
>> h = [1 -1 2 1];
          % 1<n<4
>> x = [1 2 -1 3 -2 1];
          %-1<n<4
>> y = conv(h,x)
          % 0<n<8
```

DTS Convolution

Table method:

n	-1	0	1	2	3	4	5	6	7	8
x[n]	1	2	-1	3	-2	1				
h[n]		0	1	-1	2	1				
		1	2	-1	3	-2	1			
			-1	-2	1	-3	2	-1		
				2	4	-2	6	-4	2	
					1	2	-1	3	-2	1
y[n]		**1**	**1**	**-1**	**9**	**-5**	**8**	**-2**	**0**	**1**